ExpNo 20 Experiment to understand K means clustering algorthim for a given data set

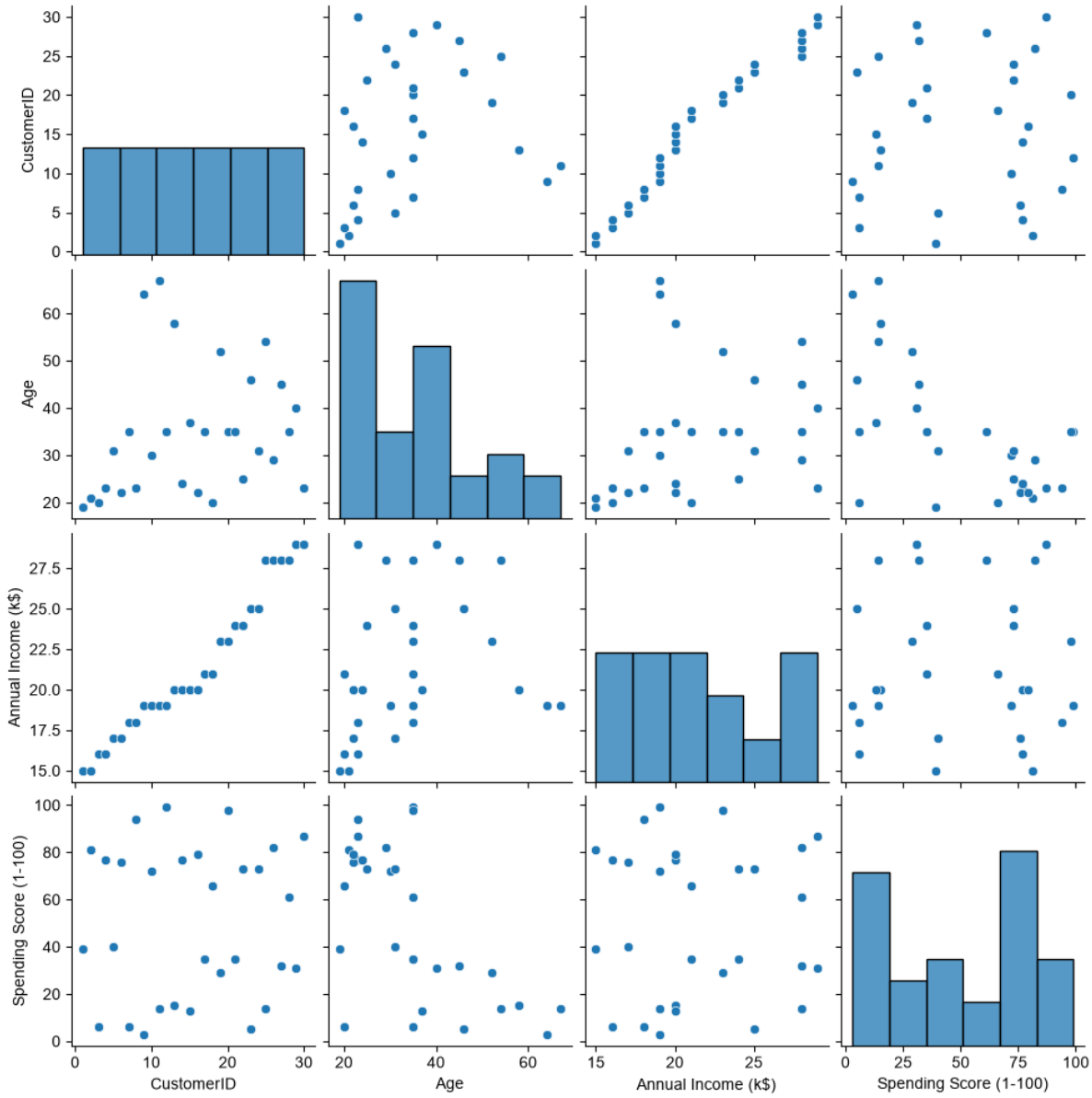
Program:

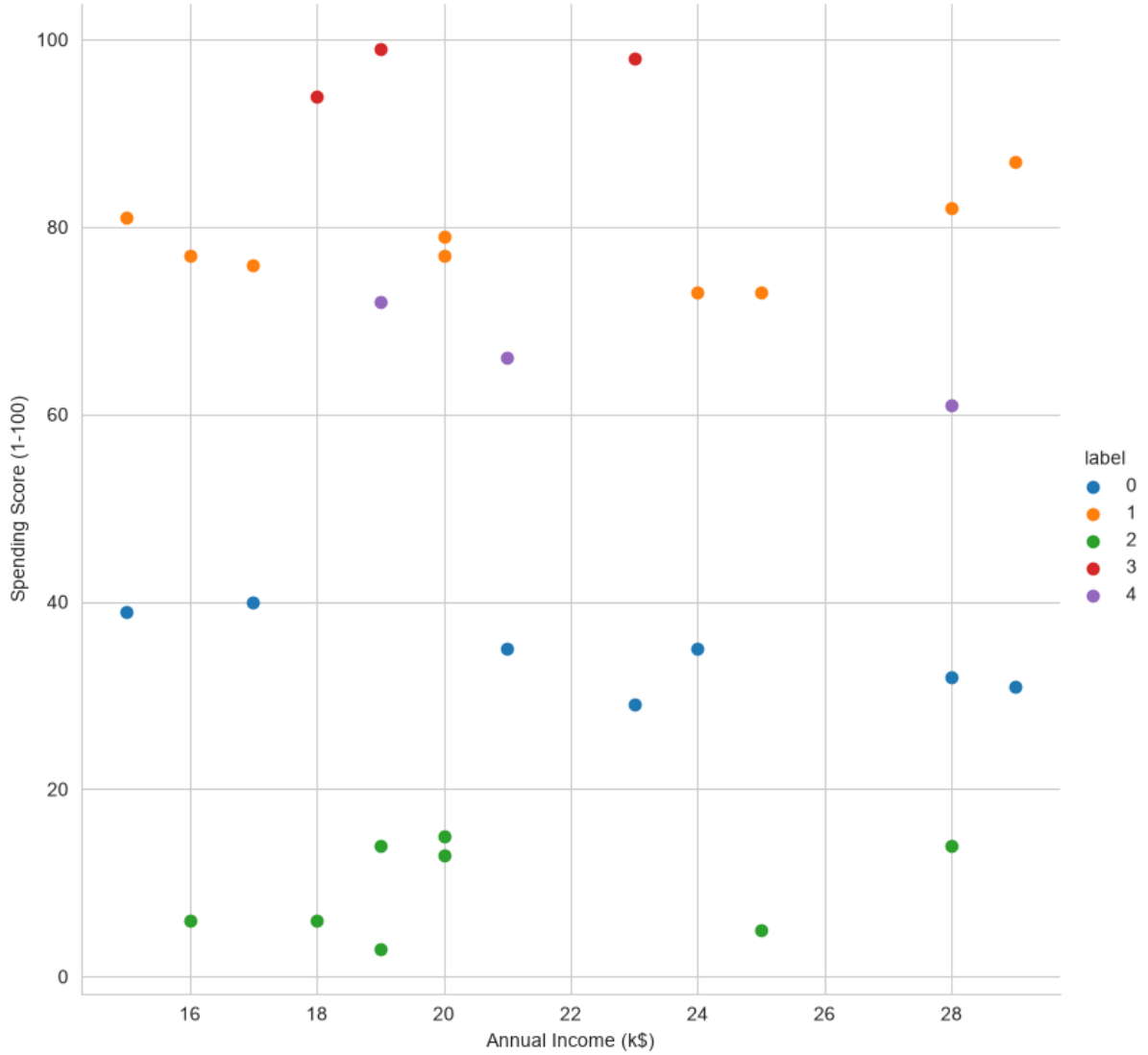
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
df=pd.read_csv('Mall_Customers.csv')
df.info()
df.head()
sns.pairplot(df)
features=df.iloc[:,[3,4]].values
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
```

```
for i in range(1,10):
  model=KMeans(n_clusters=i)
  model.fit(features_el)
  wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

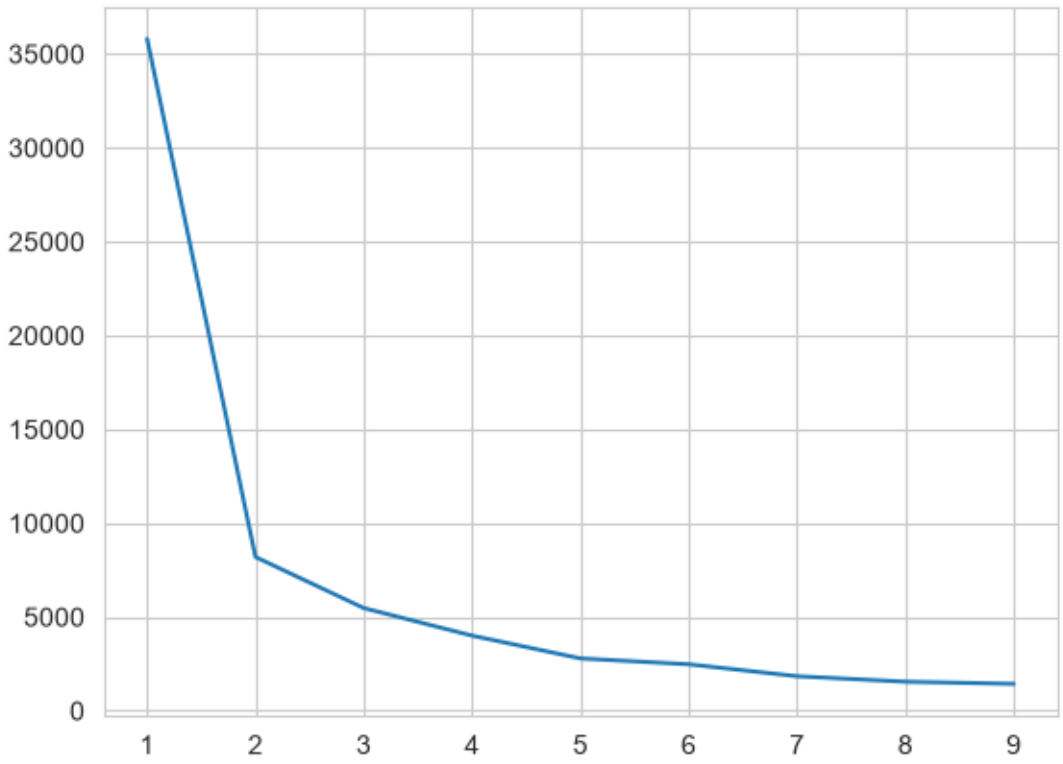
Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              30 non-null     int64
 1   Genre                   30 non-null     str
 2   Age                     30 non-null     int64
 3   Annual Income (k$)      30 non-null     int64
 4   Spending Score (1-100)  30 non-null     int64
dtypes: int64(4), str(1)
memory usage: 1.3 KB
```


CustomerID
Age
Annual Income (k$)
Spending Score (1-100)
CustomerID
Age
Annual Income (k$)
Spending Score (1-100)



[<matplotlib.lines.Line2D at 0x255899fa3c0>]


35000
30000
25000
20000
15000
10000
5000
0
1
2
3
4
5
6
7
8
9